The documentation and process conversion measures necessary to comply with this revision shall be completed by 16 October 1999.

INCH-POUND

MIL-PRF-19500/431B
16 July 1999
SUPERSEDING
MIL-S-19500/431A(USAF)
22 July 1993

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, N-CHANNEL, SILICON
TYPES 2N4091, 2N4092, AND 2N4093, JAN, JANTX, AND JANTXV

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel, junction, silicon field-effect transistors intended for use in chopper and analog gate circuit applications. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-18).

1.3 Maximum ratings. T_A = +25°C, unless otherwise specified.

Type	P_T 1/ T_A = +25°C (free air)	V_{DS}	V_{DG}	V_{GS}	I_G	T_J	T_{STG}
	W	V dc	V dc	V dc	mA d	°C	°C
2N4091	0.36	40	40	-40	10	-65 to +175	-65 to +200
2N4092	0.36	40	40	-40	10	-65 to +175	-65 to +200
2N4093	0.36	40	40	-40	10	-65 to +175	-65 to +200

1/ Derate linearly 2.4 mW/°C for T_A > +25°C.

1.4 Primary electrical characteristics. T_C = +25°C, unless otherwise specified.

Type	$r_{ds(on)}$ V_{GS} = 0 I_D = 0 f = 1 kHz		$V_{GS(off)}$ V_{DS} = 20 V dc I_D = 1.0 nA		Max I_{DSS} 1/ V_{GS} = 0 V V_{DS} = 20 V dc		V_{DS}(on) maximum		
							I_D = 6.6 mA V_{GS} = 0	I_D = 4.0 mA V_{GS} = 0	I_D = 2.5 mA V_{GS} = 0
	Ω		V dc		mA dc		V dc	V dc	V dc
	Min	Max	Min	Max	Min	Max			
2N4091		30	-5	-10	30		.20		
2N4092		50	-2	-7	15			.20	
2N4093		80	-1	-5	8				.20

1/ Pulsed (see 4.5.1).

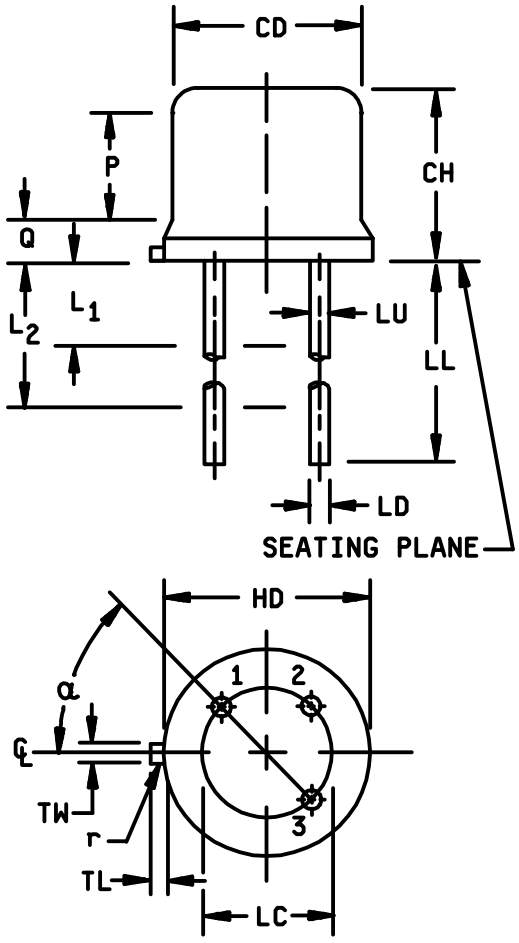
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8
LU	.016	.019	0.41	0.48	7,8
L1	---	.050	---	1.27	7,8
L2	.250	---	6.35	---	7,8
Q	---	.030	---	0.76	5
TL	.028	.048	0.71	1.22	3,4
TW	.036	.046	0.91	1.17	3
r	---	.010	---	0.25	10
α	45° TP		45° TP		6



NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TL shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 = source, lead 2 = drain, lead 3 = gate.

FIGURE 1. Physical dimensions (similar to TO-18).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (NPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and herein.

3.3.1 Lead material and finish. Lead finish shall be in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead material is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Internal construction. Multiple chip construction is not permitted to meet the requirements of this specification.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

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3.5.1 Handling. Devices must be handled with certain precautions to avoid damage due to the accumulation of electrostatic charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R ≤ 100 k, whenever bias voltage is to be applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.4).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table II of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table III herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement
	JANTX and JANTXV levels
9 , 10	Not applicable
11	I_{GSS1} , I_{DDS} , $r_{DS(on)}$
12	MIL-STD-750, method 1039, test condition A.
13	Subgroup 2 of table I herein; ΔI_{GSS} = ±50 pA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. ΔI_{DSS} = ±20 percent of initial value.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and paragraph 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection table VIb, JAN, JANTX, and JANTXV of MIL-PRF-19500).

Subgroup	Method	Condition
B2	1051	test condition G
B3	1027	Condition A, T_A = +175°C; V_{DS} = 0; V_{GS} = -24 V dc, 340 hrs.
B6	1031	T_{STG} = +200°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	test condition E.
C6	1026	V_{GS} = -24 V dc; V_{DS} = 0; T_J = +175°C.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection .

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, gate to source	3401	Bias condition C; V_{DS} = 0 V dc; I_G = -1.0 µA dc	$V_{(BR)GSS}$	-40		V dc
Gate reverse current	3411	V_{GS} = -20 V dc; V_{DS} = 0; bias condition C	I_{GSS}		-0.1	nA dc
Drain current	3413	V_{DS} = 20 V dc; bias condition C	$I_{D(off)}$		-0.1	nA dc
2N4091		V_{GS} = -12 V dc				
2N4092		V_{GS} = -8 V dc				
2N4093		V_{GS} = -6 V dc				
Drain current	3413	V_{DS} = 20 V dc; V_{GS} = 0 (pulsed, see 4.5.1); bias condition C	I_{DSS}			
2N4091				30		mA dc
2N4092				15		mA dc
2N4093				8		mA dc
Static drain to source on-state resistance	3421	V_{GS} = 0; pulsed (see 4.5.1); bias condition B; I_D = 1.0 mA dc	$r_{DS(on)}$			
2N4091					30	Ω
2N4092					50	Ω
2N4093					80	Ω
Drain to source "on" state voltage	3405	V_{GS} = 0; bias condition B	$V_{DS(on)}$			
2N4091		I_D = 6.6 mA dc			0.2	V dc
2N4092		I_D = 4.0 mA dc			0.2	V dc
2N4093		I_D = 2.5 mA dc			0.2	V dc

See footnotes at end of table.

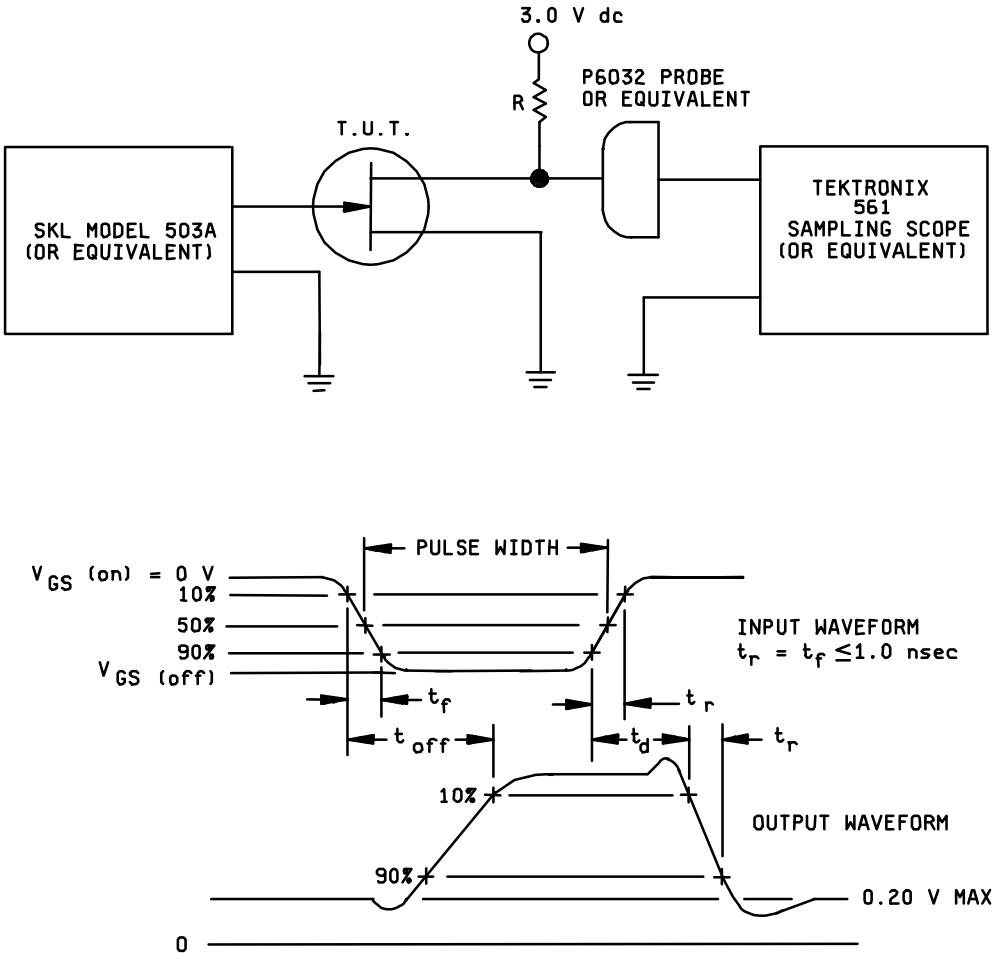
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TABLE I. Group A inspection .

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Gate current	3411	Bias condition C; V_{DS} = 0 V dc V_{GS} = -20 V dc;	I_{GSS}		-0.2	µA dc
Drain current	3413	V_{DS} = 20 V dc; Bias condition A	$I_{D(off)}$			
2N4091		V_{GS} = -12 V dc			0.2	µA dc
2N4092		V_{GS} = -8 V dc			0.2	µA dc
2N4093		V_{GS} = -6 V dc			0.2	µA dc
Subgroup 4						
Small-signal common-source short-circuit input capacitance	3431	V_{DS} = 20 V dc; V_{GS} = 0; f = 1 MHz	C_{iss}		16	pF
Small-signal common-source reverse transfer capacitance	3433	V_{DS} = 0 V dc; V_{GS} = 20; f = 1 MHz	C_{rss}		5	pF
Turn-on delay time		See figure 2	$t_{d(on)}$			
2N4091					15	nS
2N4092					15	nS
2N4093					15	nS
Rise time		See figure 2	t_r			
2N4091					10	nS
2N4092					20	nS
2N4093					40	nS
Turn-off delay time		See figure 2	$t_{d(off)}$			
2N4091					40	nS
2N4092					60	nS
2N4093					80	nS
Subgroups 5, 6, 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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NOTES:

1. Measure under pulse conditions: Pulse width ≤ 300 ms; duty cycle ± 3 percent.
2. 2N4091, I_D = 6.6 mA.
   2N4092, I_D = 4.0 mA.
   2N4093, I_D = 2.5 mA.
3. 2N4091, $I_{GS(off)}$ = -12 V.
   2N4092, $I_{GS(off)}$ = -8.0 V.
   2N4093, $I_{GS(off)}$ = -6.0 V.

FIGURE 2. Switching time test circuit and waveforms.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level and type designation.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

CONCLUDING MATERIAL

Custodian:
Air Force - 11
DLA - CC

Review activities:
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-F157)

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/431B	2. DOCUMENT DATE 99/07/16

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, N-CHANNEL, SILICON TYPES 2N4091, 2N4092, AND 2N4093, JAN, JANTX, AND JANTXV

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY				
a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510	DSN 850-0510	FAX 614-692-6939	EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAC Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC -LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvior, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888			

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